Praveen Kumar R

Data Science · Machine Learning · Deep Learning · Computer Vision

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Summary

I am an Engineering student at SSN college of Engineering, aspiring to lead my career in the research field. I am a motivated undergraduate student interested in Computer Vision and Machine Learning. I would like to work on real life application, exploiting these fields. I am good speaker with great leadership skills.

Education

SSN College of Engineering — *8.89 CGPA*

B.E. Computer Science and Engineering — *Kalavakkam*

2017 - present

The Hindu Senior Secondary School — *94.8%*

12th Standard — *Indra Nagar, Chennai*

2016 - 2017

The Hindu Senior Secondary School — *9.8 CGPA*

10th Standard — *Indra Nagar, Chennai*

2014 - 2015

Experience

Admatic Solution Pvt. Ltd., — *Adyar,Chennai-600009*

Intern — *Apr 2019 - June 2019 (2 months)*

- Admatics is a start-up company for **Computer Vision and Machine Learning**. It aims to bring automation in every aspect of daily life.
- Worked on a project called 'School of Curious'. It aims to eliminate the need for a dedicated instructor to teach simple mathematics and makes studying more interactive.
- Trained **ICR**(Intelligent Character Recognition) model for hand written character recognition. Worked on **Stereo Vision** to reconstruct of 3D models by generating point cloud images. Gained experience in various topics of computer vision and implemented it in both **Python and C++**.

Honors & Awards

2019	**Campus Ambassador**, GECF	*Global Education and Career Forum*
2019	**Merit Scholarship**, Accelerate Diversity Scholarship	*Cognizant Technology Solution*
2019	**Winner**, Java Coding Competition - National Symposium	*Madras Institute of Technology*
2019	**Winner**, Reverse Coding - National Symposium	*College of Engineering, Guindy*
2018	**Merit Scholarship**, 3rd Rank in I year	*SSN College of Engineering*
2015	**Praveen Uttarardh**, Passed with Distinction (77%)	*Dhakshin Bharath Hindi Prachar Sabha*

Languages

English · Tamil · Hindi · German

Programming Languages

C · C++ · Python · Java · XML · SQL · LaTeX · MATLAB · CSS

Skills

Deep Learning · Computer Vision · OpenCV · Tensorflow · Keras · PyTorch · Data Science · Flask · Arduino · Android Studio · Tkinter

Publications

Species Recommendation using Machine Learning - GeoLifeCLEF 2019 [paper]

CLEF CONFERENCE 2019

Number Plate Segmentation Using Deep Learning Models [submitted]

CURRENT SCIENCE

Projects

Behavioral Model: Industry 4.0 *Industry Project*

PROJECT TEAM LEADER *May 2020*

- Implemented behavioral model for Emergency Stop, Nut Runner and Vision System in a factory's production line.
- These models provide necessary feedback to a **PLC** or a process using **OPC DA protocol**.
- **Tkinter** was used to design GUI fro this project.
- Our simulation models passed all the test cases and our team secured the **first place**.

Intelligent Stock Trading Agent [link] *Mini-Project*

PROJECT TEAM MEMBER *Feb 2020*

- The objective of this project is to build an agent that can intelligently buy, sell or not perform any action with respect to a stock.
- **LSTMs, Deep Q Networks, Actor Critic methods** and **Genetic Algorithm** were used to train the agent to make decisions.

Smart Attendence Management System [link] *Smart India Hackathon - Prelims*

PROJECT TEAM MEMBER *Jan 2020*

- As a part of Smart India Hackathon, we tackled the problem statement propose by the Ministry of Youth and Affairs of India. We were asked to create a robust yet fool proof attendance management system with minimal hardware requirements.
- We created a webapp using JavaScript and an Mobile App using **REACT Native**.
- I worked on **Face verification** and **Speech Verification** part of the project.
- The actual face image was encoded to a feature vector and input image was vectorised. The **cosine distance** between these vectors was used as metrics.
- The **MFCC** features were extracted from the speech signal and a **Gaussian Mixture Model** was modeled using these vector representation of these speech signal.

K-Stack [link] *Angel Hackathon*

PROJECT TEAM MEMBER *June 2019*

- K-Stack is a web application that provides facilities to communicate with groups who share interests in same field of study.
- Users can communicate with other fellow group member by either a text chat or a real time video chat. This provides free online classes with the users mutually benefiting each other.
- My role in the project as a ML Engineer is to **automate the survey process** and to ensure the classes are rated properly. I trained a deep learning model to perform **toxicity search** over the chat and predict how good the conversation went. Based on the results the users are awarded points.
- We used Google Vision API to perform **emotion analysis** over the frames of video at constant frequency. The output of this approach was also considered during the rating process.
- Our project got good appreciation from the judging committee. We were commended for the innovation in the idea.

Automatic Number Plate Recognition System [link] *Industry Project*

PROJECT ASSISTANT *Dec 2018 - July 2019*

- APNR is a model that aims to automate the surveillance vehicle entering and exiting the factory.
- The project makes use of **U-Net** segmentation algorithm to segment the number plate from the image The model was custom trained
- The project employees three **CNN models** , one to validate the selection of number plate from the image and another for the **Optical Character Recognition** (OCR) for alphabets and another OCR for numbers to resolve the problem of inter-class similarities.

School Of Curious

Internship

Project Team Member - Intern

Apr 2019 - June 2019

- The project was aimed to make learning more interactive using Machine learning and computer vision.
- The primary motive of the project is to reduce the cost of using specialized smart boards. We achieved this by placing an affordable camera at a constant relative position to the projector and white board. With proper image processing techniques combined with well **trained deep learning models** we were able to achieve equivalent outputs.
- I also worked on **stereo images** and **generation of point cloud images**, a **3D reconstruction of 2D images**, to find the original height of a distant object without a reference scale.
- I learnt how to be systematic and document all codes properly.

GeoLifeCLEF [link]

CLEF Conference 2019

Project Team Member

Mar 2019 - May 2019

- This is a research project for a competition conducted as a part of International conference (ImageClef).
- The aim of the research is to find the occurrence of all species for the particular latitude and longitude using environmental data at that location.
- Under the guidance of Dr.C.Aravindan and Dr.P.Mirunalini we were able to successfully make 12 submissions for this task
- Out of 54 submissions made all of our submissions were **placed in top 30**, and our best submission got **6th rank**.
- Our working notes paper got a **valuation score of 3 (strong accept)**.

Flight Delay Prediction [link]

Personal

Jan 2019 - June 2109

- This is a research project that aims to predict the delay of the fight based on the weather condition and time of departure.
- There are two different dataset containing the information of flight and weather. The task is merge these two datasets and pre-process them such that it is applicable to the model used for **classification and regression** task.
- The project was successfully completed with a **validation score of 0.9441.**
- The documentation of the work can be found **[here]**.

Machine Hand: Arduino

Team Memeber

Sep 2018

- Machine hand is an Arduino project that translates the motion of human hand into a robot hand motion.
- The Data required for simulation is collected from a **IMU sensor** that is attached to a person's hand and the motion is simulated by a servo attached to the robot hand.

Air Hockey: Game in C++ [link]

Team Leader

Nov 2016 - Jan 2017

- Air Hockey is a our take on the game commonly played in many amusement places.
- The project employees C-garphics and a C++ file handling system to give absolute smoothness in the working.
- The game also provides a single player mode with 4 varying level of difficulties.

Extracurricular Activities

U&I - NGO

Chennai, India

Volunteer

July 2020 - Present

- Volunteered to teach English and basic arithmetic to underprivileged kids in the local community.
- Spent 2 hours a week teaching and engaging in fun activities with the students in the NGO.

DevFest2020 - SSN

SSN College of Engineering

Reviewer

Apr 2020 - Present

- Reviewed the contestants codes and gave suggestion to code professionally.
- I was in-charge of reviewing python automation code.

Invente:3.0 - SSN

SSN College of Engineering

Organiser

Jan 2020

- Organise a event called Code relay, a multi-disciplinary event emphasising few important disciplines of Computer Science
- I set the question paper for the ML and AI discipline for the event.

I-CELL Community - SSN

SSN College of Engineering

Speaker & Leader

Sep 2017 - Present

- Gave a series of 10 lectures on **Git**, **Machine Learning** and its mathematical frame work, **Image Processing** and **Artificial Neural Network**.
- Organised hands-on sessions for all these topics, solving real world problem statements.
- **Open source contribution** was also explained in detail.
- All reference material can be found **[here]**

Computational Thinking Workshop *SSN College of Engineering*

Organizing Committee Member & Workshop host *Aug 2018 & Aug 2019*

- Helped in conducting workshop for first year undergraduates to emphasises the importance of computational thinking.
- Hosted a 4 hour workshop session on android development using 'Thunkable'

Teach A School - SSN *SSN College of Engineering*

Student Volunteer and Teacher *Aug 2018*

- TAS is a SSN community the woks for a noble cause to give educational support for those who are in need.
- We went to nearby government schools where there are insufficient number of teachers to handle the kids and helped them.

Online courses

2018	**Machine Learning A-Z™: Hands-On Python [certificate]**, UDEMY	*Completed*
2018	**Digital Image Processing Using OpenCV python3 [certificate]**, UDEMY	*Completed*
2018	**A-Z Deep Learning using python [certificate]**, UDEMY	*Completed*
2019	**Liner Algebra: 18.06**, MIT-OCW	*Completed*
2019	**Multivaribale Calculus**, MIT-OCW	*Completed*
2019	**Machine Learning**, Coursera	*In Progress*
2020	**Neural Networks and Deep Learning [certificate]**, Coursera	*Completed*
2020	**Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization [certificate]**, Coursera	*Completed*
2020	**Structuring Machine Learning Projects [certificate]**, Coursera	*Completed*
2020	**Convolutional Neural Networks [certificate]**, Coursera	*Completed*
2020	**Sequence Models [certificate]**, Coursera	*Completed*

Research Community

Institute of Electrical and Electronics Engineers *95416197*

Member

2018 - present

Association of Computer Machinery *1749623*

Memeber

2017 - present